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Generative AI Steering and Human Authoring: Complementary Search-as-Learning Scaffolds in Cross-Session Aggregated Academic Search

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Abstract

Aggregated academic search (combining search results from multiple sources) often spans multiple sessions. End-of-session *consolidation*, consisting of organizing and synthesizing gathered information so the search can be resumed later, is important but underexplored in search-as-learning. Generative AI (GenAI) has the potential to make consolidation more efficient, but it may also shift cognitive work away from the learner. We compare two approaches: *STAIS* (Steerable AI Summaries), whose GenAI summaries searchers can correct, reorganize, and fact-check, and *NT* (Note-Taking), where searchers author their own consolidations. In a two-session between-subjects study (n = 36), *STAIS* supported faster task resumption and was associated with broader conceptual coverage and greater gains in self-reported domain knowledge, whereas *NT* was associated with richer inter-concept connections and greater gains in self-reported certainty. Rather than uniformly supporting learning, the two scaffolds shaped which aspects of post-resumption learning were more strongly supported, positioning the division of consolidation labour between system and learner as a key design consideration.

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Keywords: GenAI summarization; Cross-session search; Search task resumption; Search-as-learning; Aggregated search; Academic search

1. Introduction

Academic search rarely unfolds in a single session [1]. When returning to a previous task, searchers must reconstruct enough context to continue productively [2]. The external cues available at that moment strongly shape how quickly and how well the task can be resumed [1]. In aggregated academic search, where evidence is drawn from heterogeneous sources such as scholarly libraries and reference encyclopedias [3], this reacquaintance challenge is amplified by the volume and diversity of saved material that users must reintegrate.

When the search focuses on discovery and learning, resumption is not only a memory problem: returning to a task also requires reconnecting with what has already been learned, which makes *consolidation* a shared concern of search-

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as-learning [4] and cross-session search. GenAI cluster summaries can improve resumption efficiency [5, 6], but how delegating part of consolidation to the system affects learning after resumption remains unclear: faster reacquaintance is valuable, but not if it comes at the cost of weaker knowledge construction. Rather than asking whether GenAI simply helps or harms learning, this work examines how different allocations of consolidation work between system and learner shape which facets of post-resumption learning are most strongly supported.

2. Related work

Search-as-learning (SAL) research treats information seeking not only as retrieval, but also as a process through which people build knowledge and understanding [4]. Within this tradition, the revised taxonomy of Anderson and Krathwohl (A&K) [7] provides a useful lens for both task design and outcome interpretation because it distinguishes among knowledge types and cognitive processes. For open-ended academic search tasks, the most relevant targets often lie in *conceptual knowledge* and in the processes of *understand* and *analyze*.

When users return to a suspended task, the quality of available context cues strongly affects reacquaintance and continuation [1]. What has received less attention is whether cues *authored by the learner* differ in their effects from cues *generated by the system* [8]. That distinction may matter because human-authored representations can serve two roles: external cues for later resumption and encoding activities at the time of creation.

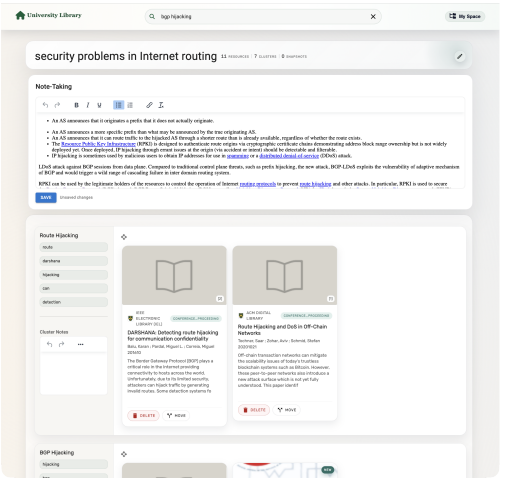
GenAI offers a new way to support consolidation by summarizing a user's saved materials [8, 9], but from a learning perspective the implications are less straightforward. The ICAP (Interactive–Constructive–Active–Passive) framework [10] predicts that learning outcomes improve as engagement moves from passive toward constructive or interactive modes. ICAP therefore implies that offloading cognitive work to the system may push engagement below the constructive level. However, Cognitive Load Theory (CLT) [11] suggests that removing extraneous reacquaintance effort may free cognitive resources for deeper integration. These predictions point to a trade-off that warrants study.

3. Search interface designs

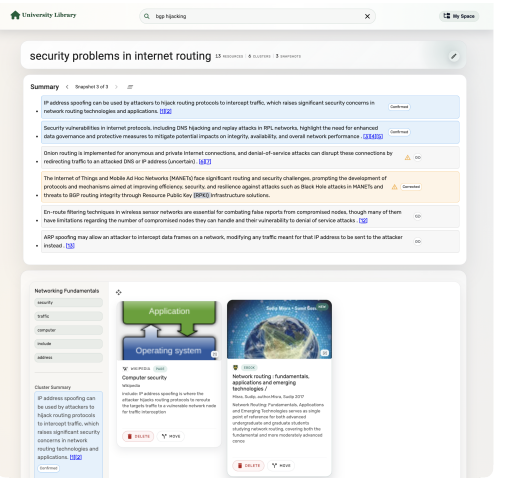
A foundation for this research is an aggregated academic search framework for cross-session searching, which our prior work [5, 12] built by extending a multi-workspace framework [13] with an aggregated-search and clustering layer. Results from our university's academic library and Wikipedia are merged on one results page, with saved resources organized into thematic clusters. A large language model (LLM; GPT-4o-mini) assigns each saved resource to the existing cluster whose accumulated content best matches it, or uses it to seed a new cluster when none fits, and gives each cluster a short keyword-style label. Users can rename, reorder, split, and merge clusters and drag resources between them. The full workspace state, including all user modifications, persists across sessions.

Note-Taking (*NT*) is the human-authored scaffold. Each cluster contains an embedded rich-text editor directly beneath its label (Figure 1a), where users can write summaries in their own words, record why resources were saved, note open questions, or leave reminders for later work. A global note area holds observations that span multiple clusters. All consolidation work in this condition is done by the learner; only the cluster labels are GenAI-authored.

Steerable AI Summaries (*STAIS*) is the user-steerable GenAI scaffold (Figure 1b). Each cluster contains a short bullet-point summary directly beneath its label, and the workspace also includes a summary spanning all clusters. These summaries are generated only from the resources saved within the relevant scope and are regenerated whenever a cluster's content changes; each bullet ends with inline citation markers linking the claim to its supporting saved resources. To reduce passive consumption and preserve user agency, *STAIS* supports three forms of steering: (i) *structural steering through reorganization*, since moving, splitting, merging, or renaming clusters changes the evidence context from which the summary is produced; (ii) *inline bullet correction*, in which users directly revise any summary bullet; and (iii) *fact-check-driven revision*, in which a fact-checking mode compares each bullet against its cited evidence and returns one of four verdicts (*Supported*, *Partial*, *Contradicted*, or *Missing Evidence*), with a suggested correction where needed, which users can accept, revise, or dismiss. User bullet edits and accepted fact-check corrections persist and are reapplied whenever a summary is regenerated. These mechanisms aim to shift interaction with GenAI summaries from simple reading toward critique, repair, and evidence-based refinement.



(a) *NT*: notes in the global note area (top) and a note editor beneath each cluster label.



(b) *STAIS*: GenAI bullet-point summaries with inline citation markers; badges and icons show fact-check and correction status.

Fig. 1: The workspace in the two interface conditions.

4. Overview of findings

In a controlled two-session between-subjects study, participants ($n = 36$) were assigned to either *NT* or *STAIS* to perform a simulated exploratory academic search task: preparing a research brief on *security problems in Internet routing*. The initial session consisted of 10 minutes of searching and 5 minutes of consolidation. One week later, participants resumed searching and finished with a writing assignment, completed with the workspace in view and scored for *concept count* (accurate, domain-relevant concepts) and *connection count* (explicit links among them). Herein, we summarize the findings; a future journal article will report the full statistics and provide detailed analysis.

Getting back to work. Saved-resource precision and interaction patterns did not differ significantly in the initial session. On return, *STAIS* participants reacquainted themselves faster, saved resources with higher precision, moved more items between clusters, and reported stronger support for task resumption. One plausible explanation for the faster re-entry lies in the scaffolds themselves: returning *STAIS* participants were presented with ready-made GenAI summaries that externalized the structure of their prior work, whereas *NT* participants were presented with their previously prepared notes, which may have been harder to consume without a condensed structure.

How the scaffolds felt. On return, *STAIS* was rated more useful, satisfying, and rewarding. *STAIS* participants reported more metacognitive strategy use in both sessions, plus more planning and progress evaluation on return, whereas *NT* participants reported more adaptation in the resumed session only. Perceived support for exploratory search split: *NT* for *uncertainty reduction* and *STAIS* for *information extraction*. Both track the consolidation mechanisms: authoring notes requires *NT* participants to articulate explicitly what they understand, which suits uncertainty reduction, while *STAIS* automates much of the extraction, which may have made information extraction feel easier.

Breadth versus depth. Concept count was higher for *STAIS*, and this advantage remained after controlling for the additional Wikipedia resources its participants saved. Connection count was higher for *NT*, plausibly because writing one's own prose invites learners to lay out how ideas relate, whereas GenAI summaries in *STAIS* were structured around extracted concepts rather than relations among them. Self-reports split the same way in both sessions: greater increases in knowledge and interest for *STAIS*, and in certainty (and, in the initial session, confidence) for *NT*.

5. Discussion and conclusion

The two interfaces studied produced different shapes of cross-session search support, rather than evidence that one approach was outright better than the other. Viewed through the lens of the A&K taxonomy [7], *NT* supported

deeper relational understanding of the material that was covered (*Conceptual/Analyze*), while *STAIS* supported broader coverage of the material itself (*Conceptual/Understand*), along with greater gains in self-reported knowledge.

A content analysis of notes and corrections showed both conditions shifting toward more cognitively demanding activities on return, but through different routes: *NT* participants shed lower-order activities, whereas *STAIS* participants added more higher-order corrections on top of a continued need to correct factual errors produced by the GenAI. The *STAIS* corrections also re-framed or elaborated on the generated content: in ICAP terms [10], participants moved beyond passive reading toward constructive engagement with the summaries. Through a CLT lens [11], the pattern is consistent with GenAI summaries freeing effort that would otherwise go into reacquaintance, and human-authored notes channelling effort into relational processing. Recent studies argue that chat-style LLM responses may support shallower learning than more effortful information seeking [14]; here, with GenAI built into the search workspace, the result was a trade-off between breadth and depth rather than an overall weakening.

The main contribution is to position the allocation of consolidation labour between learner and system as a consequential design variable in GenAI-supported search-as-learning scaffolds. The primary design implication is that consolidation support should be aligned with the learner's immediate objective rather than applied uniformly: GenAI scaffolding when conceptual mapping and rapid re-entry are the priority, and human-authored consolidation when relational integration and explanatory structuring are the goals. Because fluent prose can inflate perceived understanding [15], GenAI content should also be inspectable, source-linked, and easy to revise (as in *STAIS*) so that perceived understanding remains anchored in verified understanding.

Future work includes the design of a cross-session search-as-learning scaffold that builds upon the strengths of each approach, integrating steerable GenAI summaries with human authoring of notes. A user study will investigate whether searchers can simultaneously achieve both breadth and depth in their learning with such support.

References

[1] Y. Li, R. Capra, Stopped yet completed: Exploring the relationships between session-stopping reasons, information types, and cognitive activities in cross-session searches, in: Proceedings of the Conference on Human Information Interaction and Retrieval, 2024, pp. 119–129. doi:10.1145/3627508.3638304.

[2] J. Trafton, E. M. Altmann, D. P. Brock, F. E. Mintz, Preparing to resume an interrupted task: Effects of prospective goal encoding and retrospective rehearsal, International Journal of Human-Computer Studies 58 (5) (2003) 583–603. doi:10.1016/S1071-5819(03)00023-5.

[3] J. Arguello, Aggregated Search, Foundations and Trends® in Information Retrieval 10 (5) (2017) 365–502. doi:10.1561/1500000052.

[4] K. Urgo, J. Arguello, Search as learning, Foundations and Trends® in Information Retrieval 19 (4) (2025) 365–556. doi:10.1561/1500000084.

[5] M. Momeni, O. Hoeber, Cross-session aggregated search: Organizing and summarizing found resources, Proceedings of the Association for Information Science and Technology 62 (1) (2025) 1593–1596. doi:10.1002/pra2.1478.

[6] M. Momeni, O. Hoeber, Does it still make sense? organizing and summarizing resources in cross-session aggregated search, CHIIR '26, 2026, p. 28–39. doi:10.1145/3786304.3787873.

[7] L. W. Anderson, D. R. Krathwohl, P. W. Airasian, K. A. Cruikshank, R. E. Mayer, P. R. Pintrich, J. Raths, M. C. Wittrock, A taxonomy for learning, teaching, and assessing: A revision of Bloom's Taxonomy of Educational Objectives, Longman, New York, 2001.

[8] K. Orin, O. Hoeber, Organizing found information in public digital library search, in: Proceedings of the ACM SIGIR Conference on Human Information Interaction and Retrieval, 2025, pp. 358–366. doi:10.1145/3698204.3716471.

[9] M. H. Payandeh, J. Zhao, SenseSync: Supporting collaborative information-seeking with the involvement of large language models, in: Proceedings of the 51st Graphics Interface Conference, 2025, pp. 1–19. doi:10.1145/3769872.3769886.

[10] M. T. H. Chi, R. Wylie, The ICAP framework: Linking cognitive engagement to active learning outcomes, Educational Psychologist 49 (4) (2014) 219–243. doi:10.1080/00461520.2014.965823.

[11] J. Sweller, Element interactivity and intrinsic, extraneous, and germane cognitive load, Educational Psychology Review 22 (2) (2010) 123–138. doi:10.1007/s10648-010-9128-5.

[12] M. Momeni, O. Hoeber, A study of search result aggregation approaches for the digital humanities, Journal of the Association for Information Science and Technology 76 (11) (2025) 1488–1507. doi:10.1002/asi.70006.

[13] S. Gomes, M. Boon, O. Hoeber, A study of cross-session cross-device search within an academic digital library, in: Proceedings of the International ACM SIGIR Conference on Research and Development in Information Retrieval, 2022, pp. 384–394. doi:10.1145/3477495.3531929.

[14] S. Melumad, R. Yun, Experimental evidence of the effects of large language models versus web search on depth of learning, PNAS Nexus 4 (4) (2025) pgaf082. doi:10.1093/pnasnexus/pgaf082.

[15] A. Koriat, R. A. Bjork, Illusions of competence in monitoring one's own knowledge during study, Journal of Experimental Psychology: Learning, Memory, and Cognition 31 (2) (2005) 187–194. doi:10.1037/0278-7393.31.2.187.